PC-FT-CMPG322.1-1

Requirements for this paper/Benodighede vir hierdie vraestel:				Resources/Hulpmiddels:
Answer Scripts/ Antwoordskrifte:	X	Multi-choice cards (A4)/ Multikeusekaarte (A4)		
Attendance Slips (Fill-in Paper)/ Presensiestrokies (Invulvraestel):		Graph Paper/ GrafiekPapier		
Scrap Paper/ Rofwerkpapier	X	Calculators/ Sakrekenaars	X	
Multi-choice cards (A5)/ Multikeusekaarte (A5)		Laptop (Power not provided)/ Skootrekenaar (Krag word nie voorsien nie)		

Type of Assessment/ Tipe Assessering:	Exam Opportunity:1	Duration/ Tydsduur:	3 Hrs 00 Min
Paper Number/ Vraestel Nommer:	1	Maximum Marks/ Maksimum Punte:	100
Module Code/ Modulekode:	CMPG322	Module Description/ Module Beskrywing:	DECISION SUPPORTS SYSTEMS 2
Examiner(s)/ Eksaminator(e):	Dr A van der Merwe-PC;	Date/ Datum:	2022/11/16
		Time/ Tyd:	14:00
Internal/Interne Moderator(s):	Dr W van Blerk-VC;	Qualification/ Kwalifikasie:	BSc
External Moderator(s)/ Eksterne Moderator(s):			

Submission of answer scripts/Inhandiging van antwoordskrifte: ORDINARY

2. Mick Karra is the manager of MCZ Drilling Products, which produces a variety of specialty valves for oil field equipment. Recent activity in the oil fields has caused demand to increase drastically, and a decision has been made to open a new manufacturing facility. Three locations are being considered, and the size of the facility would not be the same in each location. Thus, overtime might be necessary at times. The following table gives the total monthly cost (in $1,000s) for each possible location under each demand possibility. The probabilities for the demand levels have been determined to be 20% for low demand, 30% for medium demand, and 50% for high demand.

	Demand is low	**Demand is medium**	**Demand is high**
Ardmore, OK	85	110	150
Sweetwater, TX	80	100	140
Lake Charles, LA	110	120	130

2.1. Which location would be selected based on the optimistic criterion?

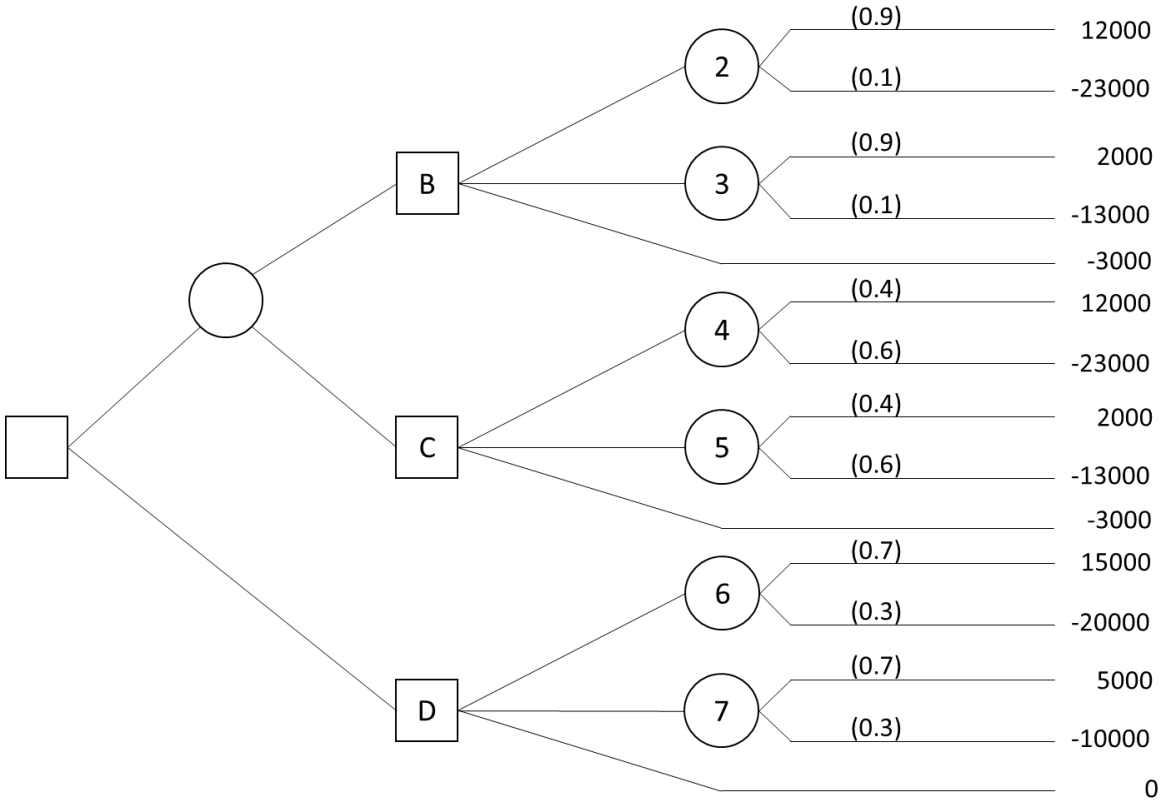
2.2. Which location would be selected based on the minimax regret criterion?

2. Mick Karra is die bestuurder van MCZ Drilling Products, wat 'n verskeidenheid spesiale kleppe vir olieveldtoerusting vervaardig. Onlangse aktiwiteit in die olievelde het die aanvraag drasties laat toeneem, en 'n besluit is geneem om 'n nuwe vervaardigingsfasiliteit te open. Drie liggings word oorweeg, en die grootte van die fasiliteit sal nie dieselfde wees op elke plek nie. Oortyd kan dus nodig wees met tye. Die volgende tabel gee die totale maandelikse koste (in $1000s) vir elke moontlike ligging onder elke aanvraagmoontlikheid. Die waarskynlikhede vir die aanvraagvlakke is bepaal as 20% vir lae aanvraag, 30% vir medium aanvraag en 50% vir hoë aanvraag.

2.1. Watter ligging sal gekies word op grond van die optimistiese kriterium? (2)

2.2. Watter ligging sal gekies word op grond van die minimax spyt kriterium? (2)

2.3. Which location should be selected to minimize the expected cost of operation?	2.3. Watter ligging moet gekies word om die verwagte bedryfskoste te minimaliseer?	(2)
2.4. How much is a perfect forecast of the demand worth?	2.4. Hoeveel is 'n perfekte voorspelling van die aanvraag werd?	(2)
2.5. What is the expected value of perfect information in this situation?	2.5. Wat is die verwagte waarde van perfekte inligting in hierdie situasie?	(4)
3. Bill Holliday is not sure what he should do. He can build a quadplex (i.e., a building with four apartments), build a duplex, gather additional information, or simply do nothing. If he gathers additional information, the results could be either favourable or unfavourable, but it would cost him $3,000 to gather the information. Bill believes that there is a 50–50 chance that the information will be favourable. If the rental market is favourable, Bill will earn $15,000 with the quadplex or $5,000 with the duplex. Bill doesn't have the financial resources to do both. With an unfavourable rental market, however, Bill could lose $20,000 with the quadplex or $10,000 with the duplex. Without gathering additional information, Bill estimates that the probability of a favourable rental market is 0.7. A favourable report from the study would increase the probability of a favourable rental market to 0.9. Furthermore, an unfavourable report from the additional information would decrease the probability of a favourable rental market to 0.4. Of course, Bill could forget all of these numbers and do nothing. Consider the decision tree provided.	3. Bill Holliday is nie seker wat hy moet doen nie. Hy kan 'n kwadpleks bou (d.w.s. 'n gebou met vier woonstelle), 'n dupleks bou, bykomende inligting insamel of eenvoudig niks doen nie. As hy bykomende inligting insamel, kan die resultate óf gunstig óf ongunstig wees, maar dit sal hom $3000 kos om die inligting in te samel. Bill glo daar is 'n 50–50 kans dat die inligting gunstig sal wees. As die huurmark gunstig is, sal Bill $15000 met die kwadpleks of $5000 met die dupleks verdien. Bill het nie die finansiële hulpbronne om albei te doen nie. Met 'n ongunstige huurmark kan Bill egter $20000 verloor met die kwadpleks of $10000 met die dupleks. Sonder om bykomende inligting in te samel, skat Bill dat die waarskynlikheid van 'n gunstige huurmark 0,7 is. 'n Gunstige verslag van die studie sal die waarskynlikheid van 'n gunstige huurmark tot 0,9 verhoog. Verder sal 'n ongunstige verslag van die bykomende inligting die waarskynlikheid van 'n gunstige huurmark tot 0,4 verlaag. Natuurlik kan Bill al hierdie nommers vergeet en niks doen nie. Oorweeg die besluitnemingsboom wat verskaf is.	



3.1. Calculate the EMV at each of the numbered nodes.

3.2. What is your advice to Bill?

3.1. Bereken die EMV by elk van die genommerde nodusse. (9)

3.2. Wat is jou raad aan Bill? (1)

4. Anne Beck recently took over a beauty supply store. Her predecessor always ordered shampoo in quantities of 100 units. Anne is reevaluating this policy. Based on her analysis, the cost to place each order is R35 and the holding cost is R8 per shampoo bottle per year. The annual demand for this product is 2500 bottles.

4.1. Calculate the optimal quantity to order.

4.2. How many orders will, consequently, be placed based on your answer in question 4.1?

4.3. Complete the following table:

4. Anne Beck het onlangs 'n skoonheids voorraadwinkel oorgeneem. Haar voorganger het altyd sjampoe in hoeveelhede van 100 eenhede bestel. Anne is besig om hierdie beleid te herevalueer. Op grond van haar ontleding is die koste om elke bestelling te plaas R35 en die houkoste is R8 per sjampoebottel per jaar. Die jaarlikse aanvraag na hierdie produk is 2500 bottels.

4.1. Bereken die optimale bestelhoeveelheid. (2)

4.2. Hoeveel bestellings sal gevolglik geplaas word op grond van jou antwoord in vraag 4.1? (2)

4.3. Voltooi die volgende tabel:

	Average inventory	**Total order cost**	**Total holding cost**	**Total inventory cost**
Current ordering policy				
Optimal ordering policy				

(8)

4.4. Should Anne change the current order policy and, if so, how much can she save?

4.4. Moet Anne die huidige bestelbeleid verander en, indien wel, hoeveel kan sy spaar? (2)

5. The table below represent a yearly demand for litres of brandy (in 1000's) collected by you for a local liquor distributor in the Vaal triangle for 2016. You, as the manager for this liquor distributor want to improve the forecast accuracy for the next year (i.e., year 12). Being a student of CMPG322 you want to apply the techniques you have studied to your data to make your forecast effort based on a scientific approach rather than haphazard approaches. You want to determine which of the selected forecasting techniques is the most accurate based on the MAD error measure criteria.

5. Die tabel hieronder verteenwoordig 'n jaarlikse aanvraag na liters brandewyn (in 1000'e) wat deur jou vir 'n plaaslike drankverspreider in die Vaaldriehoek ingesamel word vir 2016. Jy as die bestuurder van hierdie drankverspreider wil die akkuraatheid van die voorspelling vir die volgende jaar (dus jaar 12) verbeter). As student van CMPG322 wil jy tegnieke wat jy bestudeer het, op jou data toepas om jou voorspelling te maak gebaseer op 'n wetenskaplike benadering eerder as lukrake benaderings. Jy wil bepaal watter van die betrokke voorspellings tegnieke die akkuraatste is, gebaseer op die MAD-foutmaatstaf kriteria.

Year	Demand (1000s)	Year	Demand (1000s)
1	4	**7**	7
2	6	**8**	9
3	4	**9**	12
4	5	**10**	14
5	10	**11**	15
6	8	**12**	-

5.1. Do a forecast by means of a 3-year weighted moving average method, using a weight of 2 for the most recent year and a weight of 1 for both years prior to that.

5.1. Doen 'n voorspelling deur middel van 'n 3-jaar geweegde bewegende gemiddeld metode, met 'n gewig van 2 vir die mees onlangse jaar en 'n gewig van 1 vir beide jare voor dit. (6)

5.2. Use $y = 2{,}2181 + 1.0545x$ as a trend line to estimate the yearly demand.

5.2. Gebruik $y = 2{,}2181 + 1.0545x$ as 'n tendenslyn om die jaarlikse aanvraag te voorspel. (6)

5.3. Use the MAD forecast error measure to determine which forecasting technique used in questions 5.1 and 5.2 can be considered more accurate. Explain why.

5.3. Gebruik die MAD-voorspellings foutmaatstaf om te bepaal watter voorspellingstegniek wat in vrae 5.1 en 5.2 hierbo gebruik is, as meer akkuraat beskou kan word. Verduidelik hoekom. (2)

6. You are selling cold drinks from your store to the public. All customers form a single line to pay at the cashier after selecting their cold drink. These customers arrive at a rate of 4 per minute and follows a Poisson distribution. The single cashier helps these customers at 12 seconds per customer and follows an exponential distribution.

6. Jy verkoop koeldrank uit jou winkel aan die publiek. Alle klante vorm 'n enkele lyn om by die kassier te betaal nadat hulle hul koeldrank gekies het. Hierdie kliënte arriveer teen 'n tempo van 4 per minuut en volg 'n Poisson-verspreiding. Die enkele kassier help hierdie kliënte teen 12 sekondes per kliënt en volg 'n eksponensiële verspreiding.

6.1. What is the probability that there are more than two customers in the system?

6.1. Wat is die waarskynlikheid dat daar meer as twee kliënte in die stelsel is? (2)

6.2. What is the probability that the system is empty?

6.3. How long will a customer wait on average to before reaching the cashier?

6.4. If a second cashier is added and the assumption is still that customers wait in a single line and go to the first available cashier, how will the answers in questions 6.1-6.3 change?

6.2. Wat is die waarskynlikheid dat die stelsel leeg is? (1)

6.3. Hoe lank sal 'n kliënt gemiddeld wag voordat hy die kassier bereik? (2)

6.4. As 'n tweede kassier bygevoeg word en die aanname steeds is dat kliënte in 'n enkele ry wag en na die eerste beskikbare kassier gaan, hoe sal die antwoorde vir vrae 6.1-6.3 verander? (5)

7. The time between arrivals at a drive-through window of a fast-food restaurant follows the distribution given below. The service time distribution is also given in the table below. Use the random numbers provided to simulate the activity of the first five arrivals. Assume that the window opens at 11:00 a.m. and the first arrival after this is based on the first interarrival time generated.

7. Die tyd tussen aankomste by 'n deurryvenster van 'n kitskosrestaurant volg die verspreiding wat hieronder gegee word. Die dienstydverspreiding word ook in die tabel hieronder gegee. Gebruik die ewekansige getalle wat verskaf word om die aktiwiteit van die eerste vyf aankomste te simuleer. Aanvaar dat die venster om 11:00 vm. oopmaak en die eerste aankoms hierna is gebaseer op die eerste tussenaankomstyd wat gegenereer is.

Time between arrivals	Probability	Service time	Probability
1	0.2	1	0.3
2	0.3	2	0.5
3	0.3	3	0.2
4	0.2		

Random numbers for arrival: 14, 74, 27, 03
Random numbers for service times: 88, 32, 36, 24.

7.1. Set up the random number intervals for the information provided.

7.2. Simulate the four arrivals and complete the table below:

7.1. Stel die ewekansige getal intervalle vir die inligting verskaf. (7)

7.2. Simuleer die vier aankomste en voltooi die tabel hieronder:

Customer	Time of arrival	Time of service	Waiting time before service	Time of departure
1				
2				
3				
4				

(8)

8. Users for Laptops is very particular in their choices. When they upgrade loyalty plays a major role in their choice for a new laptop. However, some user may switch to a different brand. The three brands for this application are HP, Dell and Apple. Currently, users who own HP laptops will buy HP 80% of the time while the rest will switch to other brands in equal

8. Gebruikers vir skootrekenaars is baie spesifiek in hul keuses. Wanneer hulle opgradeer, speel lojaliteit 'n groot rol in hul keuse vir 'n nuwe skootrekenaar. Sommige gebruikers kan egter na 'n ander handelsmerk oorskakel. Die drie handelsmerke vir hierdie toepassing is HP, Dell en Apple. Tans sal gebruikers wat HP-

	portions. Users who own Dell will buy Dell 90% of the time while the rest will also switch to other brands in equal portions. Users who own Apple laptops will buy Apple 70% of the time while 20% will buy HP and the rest will buy Dell. Each brand has 200,000 users who plan to upgrade next year. Using Markov analysis, determine how many laptops of each brand will be purchased next year. Please show all calculations.	skootrekenaars besit, HP 80% van die tyd koop, terwyl die res in gelyke porsies na ander handelsmerke sal oorskakel. Gebruikers wat Dell besit, sal Dell 90% van die tyd koop, terwyl die res ook in gelyke porsies na ander handelsmerke sal oorskakel. Gebruikers wat Apple-skootrekenaars besit, sal Apple 70% van die tyd koop, terwyl 20% HP sal koop en die res sal Dell koop. Elke handelsmerk het 200000 gebruikers wat beplan om volgende jaar op te gradeer. Deur gebruik te maak van Markov analise, bepaal hoeveel skootrekenaars van elke handelsmerk volgende jaar gekoop sal word. Toon asseblief alle berekeninge.
		(10)

TOTAL: 100

File reference: 8.1.7.2.2